

M534H HOMEWORK– Spring 2026

Prof. Andrea R. Nahmod

•Set 1. Due date: 02/12/2026

Section 1.1: 2, 3, 4.

Note Note that an equivalent manner (that what we discussed in Lecture 1) of viewing linearity is explained in Strauss' book page 2.

Assigned background reading: Read carefully Appendices A1. A2 and A3 as we will need and use this material through the course.

Extra Problem (do not turn in): Prove the *Second Vanishing Theorem* in A.1. page 416

•Set 2. Due date: 02/17/2026

Section 1.1: 11, 12.

Section 1.2: 1, 2, 3, 4, 5, 6.

Additional Problem 1: Solve the transport equation $5u_x - 6u_y = 0$ together with the auxiliary condition that $u(x, 0) = 4x^3$.

Additional Problem 2: Solve the inhomogeneous transport equation $2u_x + 3u_y = 1$.

Assigned Reading: Read carefully Handouts 1, 2 and 3 posted in the webpage. These correspond to Examples 2, 3 and 4 in Strauss' Section 1.3